# INDIVIDUAL WEEKLY PROJECT REPORTS

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## Role: Data Collection and Data Quality
## Project Title: Flight Delay Analysis and Visualization

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### WEEK 7 – Supporting Statistical Analysis Through Data Quality Checks

During the seventh week, I focused on supporting the statistical analysis stage by reviewing the quality and suitability of the numerical variables used by the team. Correlation analysis and PCA required appropriate numerical data, so it was important to ensure that the selected fields were consistent.

I examined the importance of checking the variables before applying statistical techniques. Missing values, invalid entries, and inconsistent numerical formats could affect the calculations and produce misleading results.

I coordinated with the team members responsible for analysis to understand which variables were being used for correlation analysis and PCA. I also reviewed the relationship between

departure delay and arrival delay, which showed a strong positive correlation of approximately 0.971 in the team's analysis.

My responsibility was to support the data quality aspect of this process and ensure that the variables used were suitable for the planned techniques. I also helped identify the importance of using consistent data during analysis and visualization.

**Outcome and Learning:**

This week improved my understanding of how data quality affects statistical results. I learned that even advanced analytical techniques depend on the accuracy and consistency of the input data.